

AI Equity Research Agent

An autonomous AI agent that reads a company's SEC filings, researches it like a buy-side analyst, fact-checks its own work and writes a fund-style investment memo, in English and Spanish.

Renzo Aldana · Barcelona, Spain · October 2026

Live demo: renzo-equity-research.streamlit.app · Code: github.com/aldanar304-code/Equity-Research-Agent-

~30
SEC filings downloaded
per company

~20
research steps (tool calls)
per memo

\$0.41–0.52
API cost per memo, about
4 minutes

54
self-corrections made by
the fact-check across 4
memos

The project in one paragraph

Given a stock ticker, the agent downloads the company's recent annual and quarterly reports, press releases, proxy statement and insider trades from the SEC (about 1,000 pages), then decides for itself what to research: 10 years of financials, competitors, analyst expectations, risks, executive pay and recent news. It values the company with a discounted cash flow (DCF) in bear, base and bull scenarios, forms a Buy / Hold / Sell view with a 12-month target, and explains where it disagrees with Wall Street. It then **re-checks its own draft against the sources and corrects its mistakes** before producing a styled PDF with six charts and EU-style legal disclosures.

Results

Company	Call	12-month target	Tool calls	Cost	Self-corrections
NVIDIA (NVDA)	Hold	\$255 (+9%)	19	\$0.43	13
Microsoft (MSFT)	Hold	\$530 (+2%)	23	\$0.52	17
Amazon (AMZN)	Hold	\$255 (+1%)	23	\$0.49	10
Apple (AAPL)	Sell	\$290 (–13%)	19	\$0.41	14

Each memo also has a Spanish version (about \$0.13 each). An automatic number check confirmed that no financial figure changed in any of the four translations.

What it demonstrates

Finance
DCF valuation and sensitivity analysis · bear/base/bull scenarios · peer multiples (P/E, EV/EBITDA) · quality-of-earnings analysis · reading 10-K, 10-Q, 8-K and proxy filings · variant perception versus consensus

Technology
AI agents and tool use (Claude API) · Python · SEC EDGAR and XBRL data · search over documents (BM25) · automated testing · data visualisation · web app deployment (Streamlit)

Responsible AI and regulation: self fact-checking, cost controls, AI-generated-content disclosure, and EU MAR / MiFID II-style disclosures, including the author's own position.

1. The problem

Professional equity research is slow and expensive. Before an analyst can form a view on a company, they read hundreds of pages of filings, rebuild a multi-year financial history, compare competitors, check what Wall Street already expects and build a valuation model. For one company that's days of work.

General-purpose AI chatbots look like a shortcut, but they have three weaknesses that make them unsuitable on their own:

1. **They don't know current facts.** They answer from training data, not from today's filings and prices.
2. **They make arithmetic slips.** Language models predict text; they don't calculate.
3. **They can state unsupported claims confidently,** with no sources and no way to check them.

Goal of the project: build an AI system that does the analyst's groundwork with real, sourced data, keeps every number verifiable, costs less than a dollar per company, and produces a memo in the format a fund would actually use.

2. How it works

The key idea: **the AI is the brain, the tools are its hands.** Claude, the AI model, plans the research and writes. Small, reliable Python functions ("tools") fetch data, search documents and do the math. Because the AI acts (it looks things up and checks them) rather than answering from memory, the system is an **agent**, not a chatbot.

1 · Download

~30 SEC filings
saved locally
free · ~25 sec

2 · Research

Claude calls ~20
tools: financials,
peers, filing search,
DCF, news
~3 min · ~\$0.30

3 · Write

Fund-style memo in
a fixed 10-section
template
included above

4 · Fact-check

Re-checks every
number against
sources and
corrects
~\$0.10

5 · Render

PDF and web page,
6 charts,
disclosures, Spanish
free

The team (one file per role)

File	Role	What it does
filings.py	The librarian	Downloads SEC filings, searches them, builds the 10-year history from SEC XBRL data, summarizes insider trades, runs the DCF
data.py	Market data assistant	Prices, multiples, quarterly statements, analyst consensus and peers from Yahoo Finance
tools.py	Switchboard	The menu of tools Claude may use; routes each request to the right function
prompts.py	Job description	The analyst instructions and memo template ("never invent figures, cite every number...")
agent.py	Manager	Runs the conversation with Claude, executes tools, enforces the budget, triggers the fact-check
render.py / translate.py	Designer / translator	Charts, PDF, disclosures; Spanish translation with a number check
app.py / cli.py	Showroom / front door	The public web demo and the command (or double-click launcher) that starts a memo

Only one file, agent.py, talks to the AI. Everything that must be reliable (downloading, calculating, formatting) is ordinary code.

3. Key design decisions

Search instead of read. One company's filings total about 3 million characters, roughly 1,000 pages. Feeding all of it to the AI would cost several dollars per memo, and most of it is boilerplate. Instead the agent searches the filings, like using a book's index or Ctrl+F, and reads only the matching passages. That keeps a memo at about \$0.45.

The math runs in code, not in the AI. Claude chooses the assumptions (growth, discount rate) and explains why; Python computes the valuation exactly. This turned out to matter: in the Apple memo, the fact-check caught Claude writing that $28 \times \$9.40 \approx \290 , when the product is \$263.

A second pass that fact-checks the first. After drafting, the agent re-reads its memo, verifies every number against the tool results, fixes errors, labels opinions as opinions and lists every change in "Fact-check notes". Across four memos it made 54 corrections.

Hard cost controls. Filings are downloaded with plain code (free). Prompt caching cuts the cost of re-reading the conversation, and a hard cap (\$1.50 per memo) reserves 30% of the budget for the fact-check.

A safe public demo. The live website shows sample memos for free. To run new research, visitors use their own API key, so a public link can never generate costs for the author.

4. The finance behind each memo

Discounted cash flow (DCF)

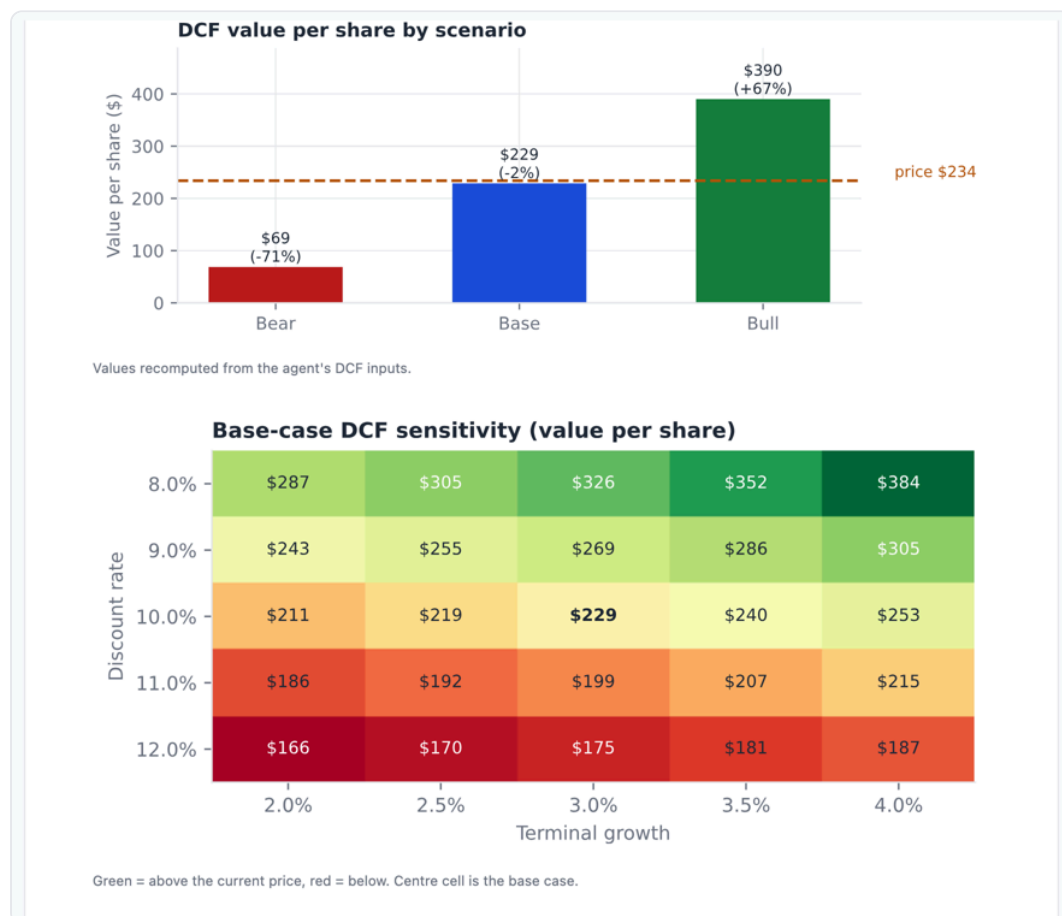
A company is worth all the cash it will produce in the future, translated into today's money. The agent starts from free cash flow (cash generated minus capital spending), projects it year by year for 8–10 years, adds a terminal value for the years after, discounts everything back at 8–11%, adjusts for cash and debt, and divides by the number of shares. A simple illustration: a business producing €100 a year forever is worth $€100 \div 10\% = €1,000$ to an investor wanting 10%, but only €500 to one wanting 20%. **A higher required return means a lower value.**

Scenarios and sensitivity

Because the future is uncertain, the DCF runs three times. For NVIDIA (price \$234):

Scenario	Assumption	Value per share	vs price
Bear	AI spending slows and cash flow falls	\$69	–71%
Base	Strong growth that gradually slows	\$229	–2%
Bull	Very strong growth for longer	\$390	+67%

A sensitivity heatmap shows how much the value moves with the assumptions: from **\$166** at a 12% discount rate to **\$384** at 8%. One number is never enough.



From the agent's NVIDIA memo: DCF value by scenario against the current price (top) and the sensitivity heatmap (bottom; green = above the price, red = below).

Valuation multiples

Like valuing a flat by its price per square metre, multiples compare the price with earnings and with similar companies. The agent picks 3–6 real competitors itself. In the NVIDIA memo, NVIDIA had the **lowest** forward P/E (14.9× versus 18.3× for Broadcom and 40.7× for AMD) despite the **fastest** growth (+106%). That's a signal either that it's cheap, or that the market expects its profits to fall later, and the memo discusses which.

Quality of earnings: is the profit real and repeatable?

Warning sign	What the agent found
One-off gains	Amazon: Q2 profit of \$62.6B included \$53.4B of gains on its Anthropic stake, so the underlying business earned far less.
Profit without cash	NVIDIA: \$59.7B of net income but only \$24.1B of operating cash in Q2, as customers took longer to pay.
Spending absorbing cash	Microsoft: record profits, but free cash flow fell two years in a row as data-centre capex reached \$116B.

Variant perception: where it disagrees with the market

Being right isn't enough; an investor has to be right about something the market has wrong. Each memo compares its view with analyst consensus. **Amazon:** 57 of 59 analysts rate it Buy with an average target of \$331 (+31%). The agent rated it **Hold at \$255**, because consensus earnings had risen 48% in 90 days largely on one-off investment gains, while free cash flow was negative and debt rising. **Apple** was the only **Sell:** at 34.8× forward earnings, none of the agent's DCF scenarios reached the current price, and its record June-quarter margin was flattered by one-off tariff refunds.

5. Proof it works: the fact-check in action

The self-check is the feature that most separates this project from a chatbot. Real corrections from the four memos:

Memo	What the agent caught in its own draft
Apple	Arithmetic error in the price target (28 × \$9.40 is \$263, not \$290). It rebuilt the target so it matched the valuation section.
Amazon	A debt ratio of 3.5× flattered by \$16B of one-off gains, corrected to an honest ~4.8×. It also reworded an FTC case described as "settled and paid", since the filing only said Amazon expected to pay.
Microsoft	Revenue growth miscalculated as 15.0% (correct: 15.7%), and a figure attributed to the wrong quarter by a web source.
NVIDIA	Cited an annual report it hadn't actually read. It replaced the citation with the documents it did use and removed the report from its "reviewed" list.

6. Real-world problems solved along the way

- **Companies change their data labels.** NVIDIA changed the label it uses for revenue in its SEC data in 2023, and reports capital spending under a different label than most companies. That first produced

empty revenue figures and overstated free cash flow. The fix merges all the alternative labels; NVIDIA's FY26 free cash flow now matches its statements (\$96.7B).

- **Messy documents.** Filings split headings across HTML tags ("RIS K FACTORS"), repeat each heading in the table of contents, and mention sections in running text. The section finder handles all three.
- **Inconsistent file names.** Earnings press releases aren't always named "ex99"; NVIDIA uses "q2fy27pr.htm". The downloader finds exhibits by what they are, not by their name.
- **Translation safety.** The number check initially flagged harmless rewording ("Q2" became "segundo trimestre"), so it was refined to focus on real financial figures. A test now verifies every chart in both languages after a translation bug was caught.

Every fix is backed by an automated test (14 in total), so problems can't silently return.

7. Responsible use and regulation

The memos are written as **general, impersonal research**, not personal investment advice, which under EU rules (MiFID II) would require authorisation from the CNMV. Because the memos contain Buy / Hold / Sell calls, they follow the EU Market Abuse Regulation's disclosure rules for investment recommendations. Every memo states that it's AI-generated, when it was produced and at what price, its methodology, the author's own positions (I hold NVIDIA shares, and this is disclosed on the NVIDIA memo), and that it's not investment advice.

8. Limitations and next steps

Limitations

- Covers only US-listed companies (SEC filings).
- Yahoo Finance data is unofficial and sometimes conflicts with filings; the agent discloses conflicts it finds.
- The fact-check reduces errors but can't guarantee there are none.
- Older EPS figures aren't adjusted for later stock splits.

Next steps

- **Public track record:** score each call after 3, 6 and 12 months, misses included.
- **Plain-English mode** for beginner investors.
- **"Ask the memo":** questions answered from the filings, with citations.
- Coverage of European companies.

9. How it was built

I designed the product and made the key decisions: the scope, the memo format, cost limits, the fact-check step, the Spanish version, the regulatory disclosures and the public-demo model. I built it with an AI coding assistant (Claude Code), tested it on real companies, and reviewed the outputs, including checking figures against the original filings. Total API spend for the whole project, including all sample memos and translations, was about \$3.

Python

Claude API (tool use)

SEC EDGAR

XBRL

Yahoo Finance

BM25 search

matplotlib

Streamlit

pytest

GitHub

Appendix: Ready-to-use text

CV: projects section

AI Equity Research Agent · Python, Claude API · renzo-equity-research.streamlit.app

- Built an autonomous AI agent that downloads a company's SEC filings and writes a fund-style investment memo: 10-year financial analysis, peer comparison, insider activity, analyst consensus and DCF valuation, delivered as a PDF with charts.
- Designed a self fact-checking step that made 54 corrections across four memos (NVIDIA, Microsoft, Amazon, Apple), plus a Spanish translation with an automatic check that every figure matches.
- Kept costs to about \$0.45 per memo through local data processing, document search and hard spending caps; added EU MiFID II / MAR-style disclosures for publishing research responsibly.

LinkedIn: Projects section

Title: AI Equity Research Agent

Description: An AI agent that does an equity analyst's groundwork. Given a ticker, it downloads the company's SEC filings (about 1,000 pages), researches 10 years of financials, competitors, analyst expectations, risks and executive pay, values the company with a DCF in bear/base/bull scenarios, and writes a fund-style investment memo with a Buy/Hold/Sell view, in English or Spanish. A second pass fact-checks the draft against its sources; across four memos it made 54 self-corrections. About \$0.45 and 4 minutes per memo. Built with Python and the Claude API.

Links: [live demo](#) and [GitHub repository](#).

LinkedIn: post

I built an AI equity research agent. 

Give it a ticker and it downloads the company's SEC filings, analyses 10 years of financials, compares it with competitors, checks insider trading and analyst expectations, runs a DCF valuation, and writes an investment memo, in English or Spanish.

My favourite part: after writing, it fact-checks itself. On Apple it caught that its own price-target math was wrong; on Amazon it noticed a debt ratio flattered by one-off gains. Across four memos it made 54 corrections to its own drafts.

Each memo costs about \$0.45 to produce. Sample memos (NVIDIA, Microsoft, Amazon, Apple): renzo-equity-research.streamlit.app

Code: github.com/aldanar304-code/Equity-Research-Agent-

Built with Python and Claude. Educational project, not investment advice.